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Analyzing and predicting city-wide electricity consumption patterns

This project focuses on analyzing and predicting electricity consumption across different city zones using data analytics and machine learning techniques. A synthetic dataset was generated containing weather conditions, special event indicators, and energy consumption records for multiple zones over one year.

The project includes:

- Data Cleaning and Pre-processing
- Exploratory Data Analysis (EDA)
- Interactive visualizations using Plotly, matplotlib
- Machine learning-based energy consumption prediction
- Interactive user input prediction system

Using:-

Python(Pandas, Numpy, Matplotlib, Seaborn, Ploty, Scikit-learn) in Jupyter Notebook

Dataset Description: -

Name	Description
Zone Id	Unique City Zone Identifier
Date	Date of Energy reading
Humidity	Daily Humidity Percentage
Special Event	Event Indicator(0-No Event,1-Event)
Avg. Temperature	Avg. daily Temp. (◦C)
Energy Consumption	Total Energy Consumed(kWh)

To Generate a Load Dataset

Generate Synthetic Dataset

The notebook automatically generates:

- 365 days of data
- 5 different city zones
- Randomized temperature, humidity, and event conditions
- Simulated energy consumption values

How to Run the Notebook

Install Required Libraries

Install dependencies using pip:

- import pandas as pd
- import numpy as np
- import matplotlib.pyplot as plt
- import seaborn as sns
- import plotly.express as px
- import plotly.graph_objects as go

Open Jupyter Notebook

Then, Open the Notebook File

City Energy Consumption Analysis & Prediction System.ipynb

Run All the Cells

Run the notebook cells sequentially to:

- Generate/load the dataset
- Perform pre-processing
- Analyze patterns
- Create visualizations
- Train machine learning models
- Predict future energy consumption

Visualizations Included

- This project includes multiple visualizations such as:
- Monthly energy consumption trends
- Correlation heatmap
- Event vs non-event energy usage comparison
- Temperature vs energy consumption scatter plots
- Interactive Plotly dashboards

Machine Learning Models Used

The following models were implemented:

- Linear Regression
- Random Forest Regressor

The models predict future electricity consumption based on:

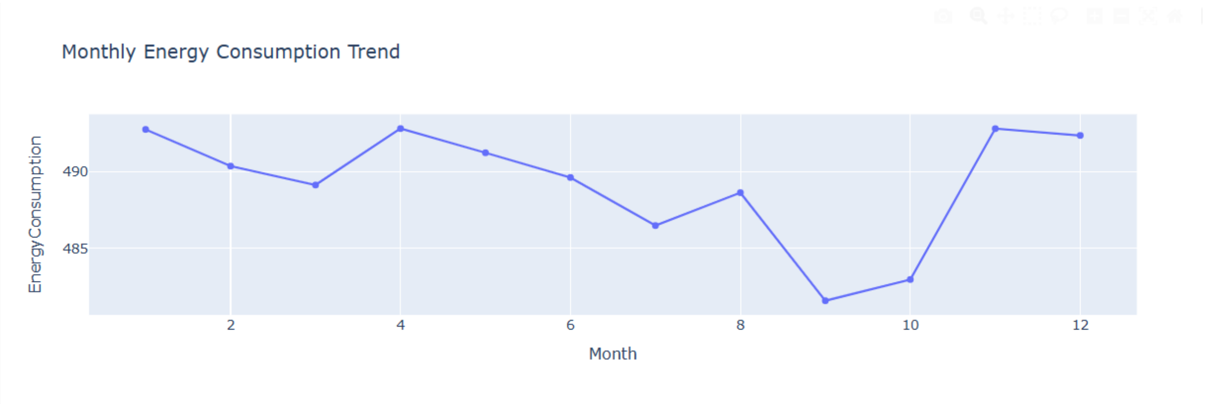
- Temperature
- Humidity
- Special events
- Zone information

Evaluation Metric:

- Mean Absolute Error (MAE)

What insights were found from the analysis?

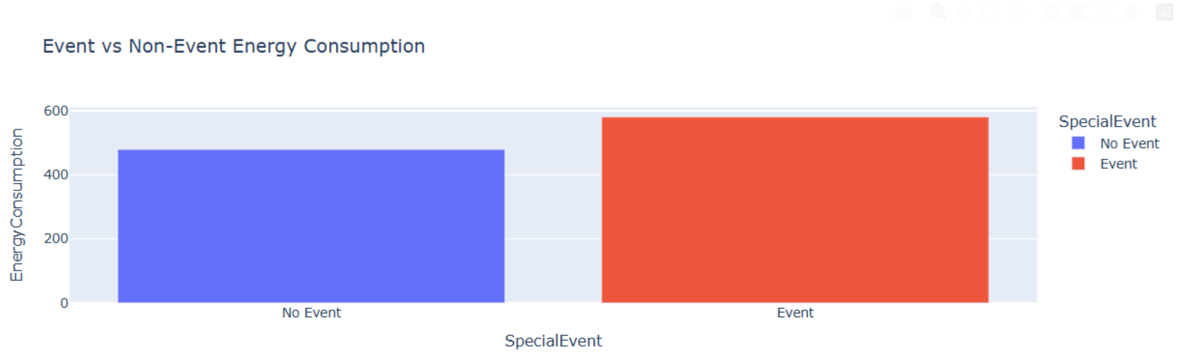
Monthly Energy Consumption Remains Stable

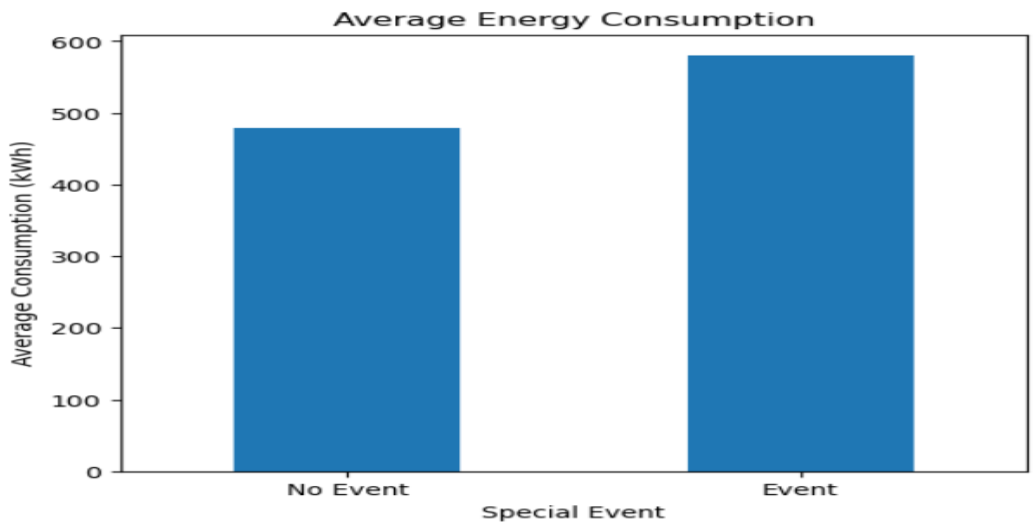


- The monthly energy consumption trend fluctuates only slightly throughout the year.
- Consumption ranges approximately between **482–493 kWh**.

- The lowest average consumption occurs around **September–October**, while the highest occurs around **April and November**.
- No strong seasonal pattern is observed, indicating relatively consistent energy demand throughout the year.

Special Events Significantly Increase Energy Usage





Energy consumption during special events is substantially higher than during normal days.

Average consumption:

- No Event: ~480 kWh
- Event: ~580 kWh

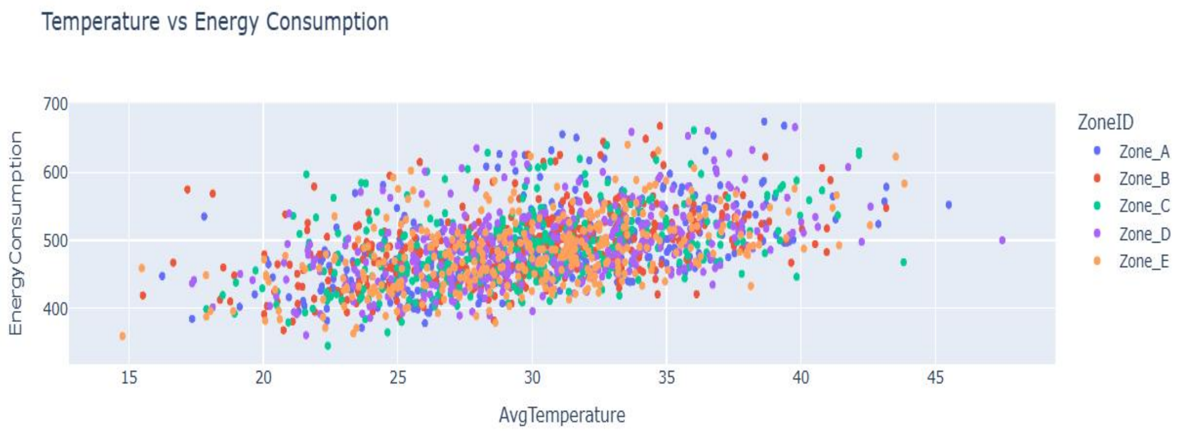
This represents an increase of roughly 20–25% during event periods.

The correlation heatmap also shows a strong positive relationship between Special Event and Energy Consumption (0.58).

Business Insight: Special events are a major driver of energy demand and should be considered when forecasting future consumption.

Festivals, sports events, and special occasions significantly increased electricity consumption across zones.

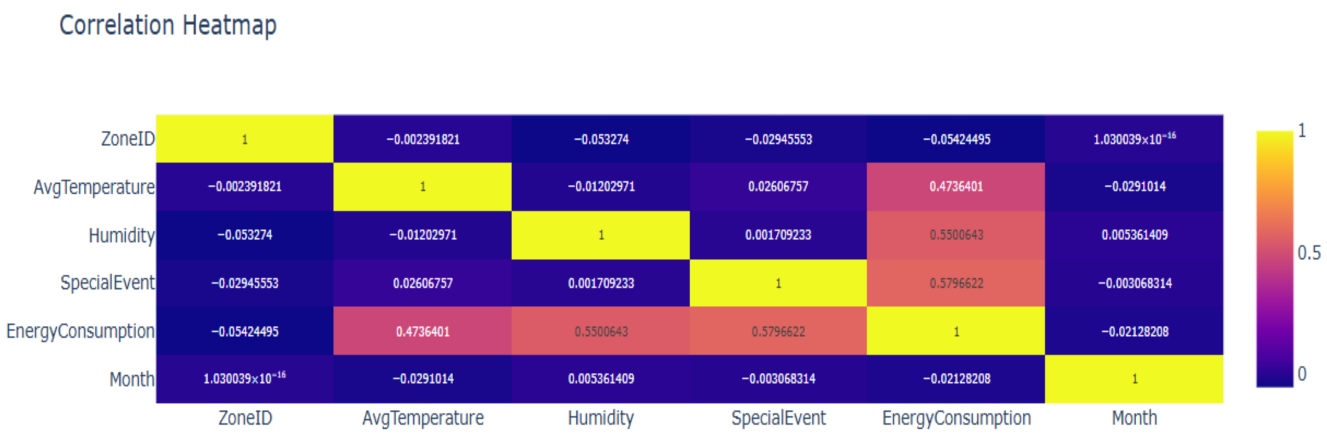
Temperature Positively Affects Energy Consumption



- ➢ The scatter plot shows that energy consumption generally rises as temperature increases.
- ➢ Correlation between **Avg. Temperature** and **Energy Consumption** is approximately **0.47**.
- ➢ Higher temperatures likely increase cooling requirements, leading to greater electricity us age.

Business Insight: Temperature is an important predictor for energy demand forecasting.

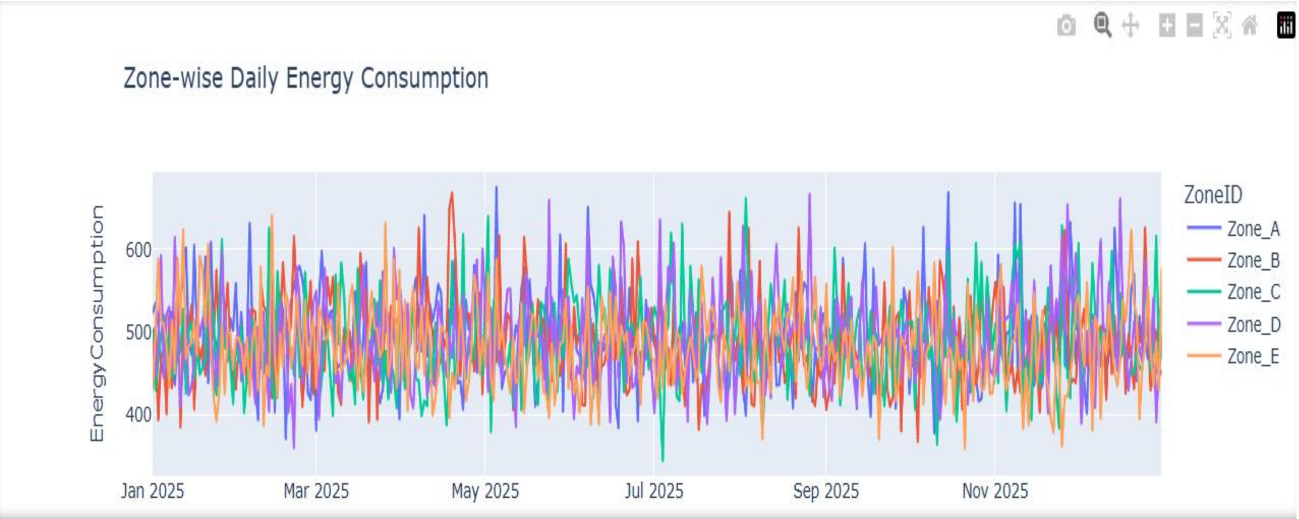
Humidity Has the Strongest Environmental Impact



- Correlation between **Humidity** and **Energy Consumption** is approximately **0.55**, stronger than temperature.
- As humidity increases, energy consumption also tends to increase.

Business Insight: Humidity should be included in predictive models because it has a significant influence on energy usage.

Zone-Wise Consumption is Relatively Similar



- Average energy consumption across all zones is close:
  - **Zone D** has the highest average consumption.
  - **Zone E** has the lowest average consumption.
- Differences between zones are relatively small, suggesting similar energy demand patterns across the city.

Business Insight: No single zone is disproportionately consuming energy, indicating balanced energy distribution.

Daily Consumption Shows High Variability

- The zone-wise daily consumption graph shows noticeable day-to-day fluctuations.
- Energy usage varies from approximately **350–680 kWh**.

- Despite fluctuations, all zones follow similar patterns, suggesting common influencing factors such as weather and events.

Month and Zone Have Minimal Direct Impact

- Correlation values:
  - **Month vs Energy Consumption:** ≈ -0.02
  - **Zone ID vs Energy Consumption:** ≈ -0.05
- These values are close to zero, indicating weak direct relationships.

Business Insight: Seasonal effects and zone location are less important than environmental conditions and special events.

Business Insights from the City Energy Consumption Analysis

1. Special Events Drive Higher Energy Demand

The analysis shows that energy consumption increases significantly during special events. Average energy usage rises from approximately **480 kWh on normal days to 580 kWh during event days**, representing an increase of around **21%**. This indicates that events are a major factor influencing electricity demand.

Business Impact: Utility providers and city planners should allocate additional energy resources during festivals, public gatherings, and special occasions to avoid supply shortages.

2. Weather Conditions Strongly Influence Energy Consumption

Both temperature and humidity exhibit positive correlations with energy usage:

- **Humidity correlation:** 0.55
- **Temperature correlation:** 0.47

Higher temperatures and humidity levels are associated with increased electricity consumption, likely due to greater use of cooling systems and air conditioning.

Business Impact: Weather forecasts can be integrated into energy demand forecasting models to improve planning and operational efficiency.

3. Energy Consumption Remains Stable Throughout the Year

The monthly trend analysis indicates relatively stable energy consumption across all months, with only minor fluctuations. No significant seasonal spikes were observed.

Business Impact: Long-term energy demand remains predictable, allowing organizations to focus more on short-term factors such as weather conditions and special events rather than seasonal adjustments.

4. Similar Consumption Patterns Across All Zones

Average energy consumption across the five zones is relatively uniform. While Zone D records slightly higher consumption and Zone E slightly lower, the differences are not substantial.

Business Impact: Energy distribution infrastructure appears balanced across the city, reducing the need for major zone-specific capacity adjustments.

5. Daily Consumption Exhibits Significant Variability

Although overall averages remain stable, daily consumption fluctuates considerably across all zones. This suggests that short-term factors influence demand more than long-term trends.

Business Impact: Real-time monitoring and predictive analytics are essential for managing sudden increases in demand and optimizing resource allocation.

6. Month and Zone Have Minimal Influence on Consumption

The correlation analysis indicates that neither the month nor the zone significantly affects energy consumption. Their correlation values with energy usage are close to zero.

Business Impact: Strategic forecasting should prioritize operational factors such as weather conditions and event schedules rather than geographic location or time of year.

<u>Conclusion</u>

The analysis reveals that **special events, humidity, and temperature are the strongest drivers of energy consumption**, while **month and geographic zone have minimal impact**. Energy demand remains generally stable throughout the year, but significant increases occur during event periods and adverse weather conditions. These findings can help improve forecasting accuracy, resource planning, and operational efficiency for city energy management.